

Presentation 2

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What did you do?

I did an analysis on repeated processes in the legal menu. First, I categorized callers who reach and also advance beyond the legal menu into four categories:

1. Listens to all the options and goes to the right menu
2. Listens to all the options, and goes to the wrong menu and goes back
 1. Indicating issue with verbiage
3. Makes a choice before listening to all the options, and goes to the right menu
4. Makes a choice before listening to all the options, and goes to the wrong menu and goes back
 1. Indicating issue with the order in which the option is being presented and / or verbiage

Additionally, I analyzed the frequency of the number of visits callers make to the legal menu, as well as a breakdown of callers making multiple visits to the legal menu (repeating the prompt, or entering a submenu and exiting back to the legal menu). Furthermore, I determined the percent of calls reaching the legal menu that were ultimately successful, defined as eventually reaching a queue.

The main part of the analysis involved identifying the percent of callers entering the wrong submenu from their first time in the legal menu (meaning they went back to the main menu) by each sub menu. I broke each of the submenu options down by whether the caller listened to the full prompt before making their choice. Finally, I showed the percent of callers listening to the full prompt before choosing a submenu (broken down by submenu). Altogether, this provides the percentages of the four types of callers for each submenu chosen.

How does it help the project?

This helps the project in identifying which submenu options have greater sources of confusion, and also with what specific issues these options may have (whether they contain verbiage issues / issue in ordering / confusion with a specific different menu / etc.) .

The analysis indicates the early options (HIV, family, housing) have the greatest rates of reaching the wrong menu (meaning they must go back to the legal menu), but the vast majority of callers choosing these submenus do not listen to the full prompt. This indicates callers may be confused a bit with the verbiage, and may also be choosing an option too early (issue with ordering and confusion/overlap with a later submenu option).

Issues faced (if any)

1. Discrepancy between the CAR data (from at least January 2025) and the flow chart. Specifically, flowchart implies routing from the main number call menu to the main menu, while CAR data implies it routes instead to the pre-legal seniors menu.
2. Additionally, an additional question surrounding how to identify callers choosing English versus Spanish at the LanguageSelectionMenu (given the English and Spanish versions of the various menu prompts have different lengths)

Attempts to resolve issues (if any)

1. Provided example contact session ID containing issue to Krish
2. To the complex EDA team's knowledge, able to directly identify the language selected by caller from the CAR data. Our compromise was to choose a length of time close to the full English and Spanish prompt length (for example, we used 120 seconds for the LegalMenu2 when the complete prompt length was 129 seconds for English and 157 seconds for Spanish). We determined that this was an appropriate compromise for the sake of analysis, given the callers with a call length between 120-129 seconds (for English speakers) and 120-157 seconds (for Spanish speakers) have effectively listened to the entire prompt (especially if the caller eventually chooses an earlier option, their intention is clear to listen to most/all of the options).

Issues resolved (if any)

Waiting to resolve issues

Next steps

- Early options (HIV, Family, Housing) have disproportionate number of revisits to legal menu, despite most callers not listening to full set of options (likely verbiage issue and also an issue with ordering / confusion with a later submenu option)
 - **Next goal #1:** Analyze the submenus chosen on second / third visit to legal menu
 - Do callers confuse two sub-menus?
 - Do callers choose an option too early?

- Which submenus do callers choose after directly repeating the legal menu prompt?
- In addition to ‘% right submenu’ and ‘% listening to full prompt’ as variables, introducing third variable (‘% eventually reaching queue’, i.e. successful call journey)
 - **Next Goal #2:** Analyze impact of multiple visits to legal menu on eventual call success
 - High percentage reaching queue after second legal menu prompt (especially for early options), may indicate the potential effectiveness of verbiage improvement
 - I.e. whether the callers initially reaching wrong sub-menu can be matched to their correct sub-menu (or the caller’s issue does not apply to any sub-menu / legal aid cannot help)

References

N/A

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Presentation 1

Link to presentation

[Link](#)

What did you do?

Our group (Will, Vince, YiHyun) collectively defined redundancies of the call journeys and partitioned them into three types: calls taking longer than they should, repeated processes, and coding/internal redundancies. For each redundancy, we identified different subclassifications of the given redundancy, and the outline for an algorithm to systematize the locating of these redundancies. I focused mostly on the third redundancy (coding / internal redundancies), identifying a redundant checking of the main hours in the main call menu when a caller does not choose a language (and is prompted to the front desk rather than the main menu). Additionally, I helped with the brainstorming of the different subtypes of repeated process redundancies (type 2) and their respective algorithms.

How does it help the project?

This brainstorming of defining/classifying different categories of redundancies serves as a solid framework, where we can utilize our algorithms to more systematically identify particular locations in the call journey that provide either an unnecessary long or confusing experience for the caller.

Issues faced (if any)

Discrepancy between the CAR data (from at least January 2025) and the flow chart. Specifically, flowchart implies routing from the main number call menu to the main menu, while CAR data implies it routes instead to the pre-legal seniors menu.

Attempts to resolve issues (if any)

Provided example contact session ID containing issue to Krish

Issues resolved (if any)

Waiting to resolve issue

Next steps

Type 3 redundancy:

- Continue flowchart analysis and identify moments where certain business hours can be consolidated, and moments where holidays are checked for a second time.

Type 2 redundancy:

- Focus on each of the four subtypes previously defined (visit to the same node multiple times, move to the previous menu -> return to the same node, move to the previous menu -> return to different node, cycles). For each, implement/code our algorithm framework/intuitions to identify the nodes with the greatest occurrence of each type 2 redundancy subtype (as well as the greatest rate of redundancy, specifically the percent of call journeys reaching a given node that are experiencing this particular redundancy subtype).

Type 1 redundancy:

- Same analysis as described in type 2; however, I believe Vince will be mainly looking at these type 1 redundancies (where calls are taking longer than they should)

References

- N/A